

## 1. Setup

The setup phase involves preparing the environment to work with D3.js and build reusable chart components. It includes installing D3.js, organizing project files, and setting up basic HTML, CSS, and JavaScript structure. Developers define common chart configurations such as width, height, margins, and scales so that charts can be reused with different datasets. This stage ensures consistency and flexibility across visualizations.

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## 2. Deployment

Deployment focuses on making the D3-based chart library usable in real applications. Charts are packaged as reusable modules or functions that can be imported into web projects. The library is tested for compatibility across browsers and optimized for performance. Deployment may involve hosting the library on a server, using version control, or integrating it with frameworks like React or Angular to enable easy reuse.

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## 3. Chart Generation

Chart generation is the process of creating visualizations using the reusable D3 components. Data is passed dynamically to chart functions, which automatically generate charts such as bar charts, line charts, or pie charts. D3 handles data binding, scaling, axes creation, and rendering. Reusability allows the same chart code to produce multiple visualizations by simply changing the data or configuration parameters.

In D3, **setup** involves including the D3 library, defining the SVG canvas size, and preparing variables such as width, height, and dataset. This step ensures that the webpage is ready to draw visual elements. **Deployment** refers to the actual creation of the chart by binding data to SVG elements (like rectangles for a bar chart), generating those elements dynamically, and positioning and styling them so the chart is rendered correctly in the browser. Together, setup and deployment convert raw data into a visible chart.

```
<!DOCTYPE html>
<html>
<head>
  <title>D3 Bar Chart</title>
  <script src="d3.v7.min.js"></script>
</head>
<body>
```

```
<script>

  var w = 400;
  var h = 200;
  var dataset = [30, 80, 45, 60, 20];

  var svg = d3.select("body")
    .append("svg")
    .attr("width", w)
    .attr("height", h);

  svg.selectAll("rect")
    .data(dataset)
    .enter()
    .append("rect")
    .attr("x", function(d, i) {
      return i * 70;
    })
    .attr("y", function(d) {
      return h - d;
    })
    .attr("width", 50)
    .attr("height", function(d) {
      return d;
    })
    .attr("fill", "steelblue");
</script>

</body>
</html>
```

OUTPUT:



◆ **How this example fits the concept**

- **Setup:**
  - D3 library is loaded
  - SVG canvas is created
  - Width, height, and dataset are defined
- **Deployment:**
  - Data is bound to rectangles
  - Bars are generated dynamically
  - Heights and positions are based on data
  - Chart is rendered in the browser